
Michael Murray

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📍 San Francisco, CA

Portfolio & More: <https://mjm-murray.github.io/>

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SUMMARY

Mechanical Engineer with 15+ years of end-to-end product development experience spanning medical devices, industrial power, and automation – from concept and prototyping through production. Holds three U.S. patents in electromechanical design. Deep hands-on expertise in DFM, FEA, GD&T, and rapid prototyping (CNC, 3D printing, injection molding), with a track record of delivering compliant, cost-effective hardware across FDA, IEC 60601, ISO 13485, and ANSI/ISO environments. Experienced leading cross-functional teams and mentoring junior engineers.

PROFESSIONAL EXPERIENCE

Nevados Engineering

Oakland, CA

Senior Mechanical Engineer

April 2025 – Present

- Redesigned the tracker cradle assembly, improving structural performance and manufacturability.
- Performed FEA on coupler components, torque tube, and fastener systems to validate structural integrity and factor of safety.
- Conducted DFMEA of the coupler assembly to identify and mitigate failure modes prior to production.
- Led coatings analysis and testing program, evaluating performance and durability across field conditions.
- Implemented and standardized SOLIDWORKS workflows, improving consistency and team efficiency.

Swope Design Solutions

San Francisco, CA

Senior Mechanical Engineer

June 2022 – April 2025

- Led a project to design and build a pump-to-catheter system for removing arterial calcified lesions.
- Created prototypes to establish proof of concept through CNC machining, molding, and 3D printing.
- Developed a positioning mechanism for a non-invasive surgical procedure to treat kidney stones.
- Built an automated data matrix scanning system, coding vision and analysis logic in C and Python.
- Spearheaded heater system design for a medical device, conducting thermal analysis and testing.
- Managed injection-molded part design and tooling, machined molds, and molded parts in silicone.
- Redesigned endoscopic robot components for injection molding (mass production), incorporating DFM.
- Mentored junior engineers and ensured alignment with project milestones.

True Digital Surgery

Santa Barbara, CA

Senior Staff Mechanical Design Engineer

November 2020 – June 2022

- Designed mechanical subsystems of a positioning system for a robotic neurosurgical 3D microscope.
- Analyzed and prevented mechanical failures and confirmed factor of safety per IEC 60601 requirements.
- Performed thermal analysis to maintain mechanical integrity under heat application.
- Developed fit/function checks using GD&T in accordance with ASME Y14.5 standards.
- Co-authored QCIC system structure per ISO 13485 guidelines.

Surround Medical Systems

Morrisville, NC

Senior Mechanical Engineer

May 2019 – May 2020

- Led the mechanical department to design a 3D x-ray positioning system from concept through production.
- Improved delivery of x-rays to reduce radiation exposure to patients and ensure image quality.
- Co-authored a 510(k) submission for FDA approval.
- Prevented mechanical failures through Finite Element Analysis (FEA) and Failure Mode and Effects Analysis (FMEA) studies.

ABB

Raleigh, NC

Senior Mechanical Engineer, R&D

March 2017 – May 2019

- Standardized transformer designs using common parts, reducing cost and lead times.
- Enhanced mechanical designs of transformers/enclosures to increase convection efficiency.
- Led a project to automate creation of drawings, models, and data, writing a standard for programming.
- Developed an outdoor-rated transformer cabinet incorporating weatherproof coating.

Ingersoll Rand	Charlotte Area, NC
Design Engineer III	August 2014 – March 2017
- Designed parts/assemblies for complex, custom industrial centrifugal air compressors (2,000–7,000 cfm, 400–1,250 HP) to meet ANSI and ISO standards. - Designed dual oil cooler system for 5000-cfm air compressors. - Optimized airflow through pre-build assessments, improving compressor performance.	
Newell Rubbermaid – Piedmont Hardware Brands	Charlotte Area, NC
Mechanical Design Engineer	February 2013 – August 2014
- Designed high-volume sheet metal, die cast, injection-molded, and stamped parts (metal and plastic). - Utilized hand calcs and SOLIDWORKS Simulation for Finite Element Analysis, guiding new product designs. - Prevented failure modes through Computer-Aided Engineering (CAE).	
Sandvik	Mebane, NC
Manufacturing Engineer	September 2011 – February 2013
- Generated CAD models, CNC programs through CAM, and drawings for high-volume metal-cutting tools. - Reduced tooling costs by 30% via optimized CAM programs for 5-axis milling machines, lathes, and grinders. - Designed and implemented a coolant reclamation system to optimize machining efficiency and cut waste.	
ABB	Lake Mary, FL
Mechanical Design Engineer	August 2009 – September 2011
- Invented a remotely operated device for industrial medium-voltage circuit breakers (5–15 kV); recognized with two patents (U.S. Patents 9,876,335 and 8,654,513B2). - Developed a Modular Mechanism Operated Cell to retrofit circuit breakers (Patent 8,878,088). - Improved circuit breaker thermal flow to prevent overheating and brazing issues.	

TECHNICAL SKILLS

DFM, DFA, DFx , (Design for Manufacturing and Assembly)	Advanced	15 yrs
CAD , (Computer-Aided Design), Drafting <i>SOLIDWORKS, Creo Parametric, (Pro/E), NX, Autodesk Inventor, Onshape, Fusion</i>	Advanced	16 yrs
Product Development, Project Management, Jira, Confluence, and Others	Advanced	10 yrs
FEA , (Finite Element Analysis)	Advanced	9 yrs
GD&T , (Geometric Dimensioning & Tolerancing), Drafting, ASME Y14.5	Advanced	9 yrs
FMEA , (Failure Mode and Effects Analysis)	Advanced	7 yrs
PDM, PLM , Software and Administration, <i>SOLIDWORKS PDM, Arena, Windchill</i>	Advanced	6 yrs
Rapid Prototyping • 3D Printing, Programming : FDM, SLA, DMLS, SLS, Maintenance & Improvement of Machines • Assembly and use of tools and equipment to make and modify parts (full list on website)	Advanced	10 yrs
Vendor Management	Advanced	8 yrs
Thermal Analysis	Intermediate	4 yrs
Computer-Aided Engineering (<i>SOLIDWORKS Simulation, Flow</i>)	Intermediate	6 yrs
CAM, (Computer-Aided Manufacturing), (<i>HSMWorks, NX</i>)	Intermediate	4 yrs
Regulations & Standards: FDA, IEC , IEC 60601, ASME, ANSI, ISO (13485 & More) , UL	Intermediate	4 yrs
Injection Molding, Plastics	Intermediate	5 yrs

EDUCATION

North Carolina State University, Mechanical Engineering, 2009
Honors: Cum Laude, Dean's List, 2008 NCSU ME Design Award, Co-Op Program

ADDITIONAL NOTES

Portfolio, Manufacturing Process Experience, and more: <https://mjm-murray.github.io/>